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Well prestige Sdn Bhd

Pengarah Urusan
Toshiba Transmission & Distribution Systems Asia Sdn. Bhd.

Tuan/Puan,

PEMAKLUMAN DAN KEPERLUAN PEMATUHAN KRITERIA TEKNIKAL BAGI *CONNECTION SPECIFICATION OF ONLINE FEEDER PILLAR FOR TNB DISTRIBUTION NETWORK* UNTUK PEMATUHAN *DIN TYPE LV OUTDOOR FEEDER PILLAR WITH REAL-TIME LOAD READING FACILITIES 1600 AMPS* (KEJ08153:2017) DAN 800 AMPS (KEJ08154:2017)

Dengan segala hormatnya perkara di atas adalah dirujuk.

2. Adalah dimaklumkan bahawa setiap pengeluar *DIN Type LV Outdoor Feeder Pillar With Real-Time Load Reading* (RTLR) *Facilities 1600 Amps (KEJ08153:2017)* dan *800 Amps (KEJ08154:2017)* diminta untuk mematuhi kriteria-kriteria teknikal seperti Lampiran 1.

3. Keperluan ini adalah untuk memastikan semua data-data yang direkodkan daripada *feeder pillar* RTLR ini dapat dihantar ke sistem ILMAS 2.0 yang berada di Bahagian Distribution Network agar tindakan analisa operasi yang berkaitan dapat dilaksanakan secara menyeluruh serta efektif. Pihak tuan boleh berhubung dengan wakil kami; Ir. Mohd Jufri bin Ismail di talian 019 - 240 6438 sekiranya memerlukan maklumat lanjut berkaitan keperluan pematuhan kepada kriteria teknikal yang dinyatakan ini.

Sekian, terima kasih.

"BETTER, BRIGHTER."

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LAMPIRAN 1

TECHNICAL SPECIFICATION FOR CONNECTION OF ONLINE FEEDER PILLAR FOR TNB DISTRIBUTION NETWORK

2023

Abbreviation		
HTTP	:	Hypertext Transfer Protocol
HTTPS	:	Hypertext Transfer Protocol Secure
IDS	:	Intrusion Detection System
IMEI	:	International Mobile Equipment Identity
IMSI	:	International Mobile Subscriber Identity
IoT	:	Internet-of-Things
IPS	:	Intrusion Prevention System
MQTT	:	Message Queuing Telemetry Transport
OFP	:	Online Feeder Pillar
PKI	:	Public Key Infrastructure
UID	:	Unique Identifiers
UIoT	:	Utility Internet-of-Things
UTF	:	UCS (Unicode) Transformation Format

Glossary

IoT	:	The connection of various devices, such as sensors, appliances, and other electronic devices, to the internet and to each other, allowing them to exchange data and perform certain functions without human intervention.
Feeder Pillar	:	An electrical distribution panel that is used to distribute electricity from a high voltage source to multiple low voltage circuits. Feeder pillars are an important component of electrical distribution systems, as they provide a safe and reliable way to distribute power to multiple locations from a single source.
Data Transmission Interval	:	"Data Transmission Interval" or simply the "Transmission Interval" is the time duration between two consecutive data transmissions from an IoT device or sensor to the cloud or other backend system. The transmission interval is typically set based on the specific application and use case, considering factors such as the required level of monitoring and control, the power consumption of the devices, and the available bandwidth and storage capacity of the backend system.
Data Logging Interval	:	"Data Logging Interval" or simply the "Logging Interval" is period during which data is collected and stored before being sent to a central system is often referred to as the This interval is typically set to a fixed duration, such as every 1 minute, 15 minutes or every 1 hour, and determines how often the data is sampled, recorded, and stored in a local memory or storage device. The logging interval is typically set based on the specific application and use case, considering factors such as the required level of detail and accuracy, the power consumption of the device, and the available memory and storage capacity.
Hardware Designed Grade	:	Refer to electronic devices and equipment designed for different purposes and operating conditions. i. Industrial-grade hardware is specifically designed and manufactured to withstand harsh operating conditions in industrial environments. These environments can include extreme temperatures, humidity, dust, vibration, and electromagnetic interference (EMI). Industrial-grade hardware typically has a ruggedized design that includes features such as heavy-duty metal casing, shock-resistant components, and protective coatings to prevent corrosion and damage from external factors. ii. Commercial-grade hardware is designed for use in business environments, such as offices, retail stores, and hospitals. Commercial-grade hardware may have features such as extended warranties, advanced security features, and remote

		management capabilities to meet the needs of business users.
	iii.	Consumer-grade hardware is designed for personal use and is typically sold through consumer retail channels. Consumer-grade hardware is often less expensive than commercial or industrial-grade hardware and may not have the same level of durability or advanced features. However, it is designed to be user-friendly and easy to set up and use.
	iv.	Entry-Level grade hardware or "budget" products is designed for consumers who are looking for a low-cost option and are willing to compromise on quality, reliability, or performance. These products are typically designed for consumers who are looking for the lowest possible cost, and may not have the same level of quality, reliability, or performance as consumer-grade hardware.

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1 Purpose

This specification defines the general technical requirement for the connection of Online Feeder Pillar (OFP), to be installed in Distribution Network (DN) of Tenaga Nasional Berhad (TNB).

This document is constructed in general to allow implementation in different projects. This scope of this specification is applicable to following types of services:

- Connection Specification to IoT Platform
- Cyber Security of Feeder Pillar

2 Scope of Work

The entire OFP will be interconnected to TNB's distribution network at the point of Unified DN UIoT Platform. The activated OFP will be connected and transferred data to the Unified DN UIoT Platform daily or real-time data transfer before sending data to TNB ICT Back-end Application Server.

2.1 Terminology	
IoT	Internet-of-Things
OFP	Online Feeder Pillar
Owner	Procuring entity, nominally an electrical distribution utility
Contractor	Supplier of the SYSTEM

3 Conformance

This Section lists relevant codes and standards that are or may be applicable to a specific system, without regard to actual project specifics such as geographical location, and so on. In some cases, there may be discrepancies for which the Owner may permit certain code deviations to Contractor. The Contractor shall make a best effort attempts to meet all applicable requirements of the latest version in this Section.

3.1 IoT Conformance	
IEEE 2413	IEEE Standard for an Architectural Framework for the Internet of Things (IoT)
ISO/IEC 30141	Internet of Things (IoT) — Reference Architecture
3.2 Cyber Security Conformance	
NIST SP 800	National Institute of Standards and Technology Special Publication
ISA/IEC 62443	Industrial communication networks - IT security for networks and systems
ISO/IEC 27001	Information Security Management

4 Connection Requirements	
4.1 Daily Data Transfer	
Communication Protocol	• SFTP • MQTT
Data Standards	• CSV
Data Logging Interval	• 15 minutes
Data Transmission Interval	• Daily (0100-0200)
Data Parameter	Refer Appendix A
Data CSV Format	Delimiter • Tab Quote • " (Double Quote) Encoding • UTF-16LE (Little Endian)
4.2 Real Time Data Transfer	
Communication Protocol	• MQTT
Data Standard	• JSON
Data Logging Interval	• 15 minutes
Data Transmission Interval	Configurable: • 1 minute • 15 minutes
Data Parameter	Refer Appendix A
MQTT Server	MQTT Broker • 175.139.190.201:1883 Topic • iot/fp/<Device UID>

5 Cyber Security Requirements	
Password Requirement	SHALL use combining of the following group: • Upper Case • Lower Case • Number • Special Character
NTP Server	SHALL Connect with NTP Server. • Option 1: TNB NTP Server • Option 2: Telco NTP Server • Option 3: Public NTP Server (https://www.pool.ntp.org/zone/my).
Device Identity	SHALL have Unique Device Identification for each device
Device Authentication	SHALL implement Password (based on password requirement)
Device-Platform Authentication	SHALL implement: • Password (based on password requirement) • Symmetric Key

	• X.509
Physical Attacks	SHALL place the device is safe place in the OFP to avoid any physical attack or tampering
IMEI Lock	SHALL disable IMEI lock if not activate for certain period due. The IMEI lock option allows users to lock the use of a SIM specifically to a device, indicated by its IMEI number. That way, even if an attacker has gained access to your device, there is no possibility to misuse your SIM by inserting it into another device.
Port & Services	SHOULD disable unneeded port & services
Firmware	SHOULD allow device to perform firmware upgrade
Intrusion Detection System & Intrusion Prevention System (IDS & IPS)	SHOULD install IDS & IPS

6 Performance Requirements	
6.1 End-of-life	At minimum, the SYSTEM end-of-life shall not be less than 10 years from the date of commissioning. It is defined when the OFP capability can no longer meet the operating requirement
6.2 Reliability	99% - that is OFP capability to start in 99 out of 100 attempts.
6.3 Availability	>99% (Annually) Where it takes following definitions: i. *Availability* is the percentage of hours that the systems is available during the year, begin upon facility commissioning. ii. In case for Real-Time Data Transfer, the system required to send the data periodically based on requirement. However, network issues due to infrastructure (cellular, fibre, etc.) are excluded. iii. In case for Daily Batch Data Transfer, the system required to record. However, planned outages for SYSTEM maintenance are excluded.

7 Applicable Issues and Supporting Control Functions	
Network Issue(s)	The SYSTEM unable to connect to IoT Platform or be connected by user/IoT Platform for a specific duration at least once a week (Sunday). Supporting Control Function(s): • Auto-restart network connection(s) • Auto-restart modem

Hardware Issues(s)	The SYSTEM shall be capable of operating based on the OFP requirement continuously. Supporting Control Function(s): • Restart hardware (if issue occurred)

Appendix A: OFP Parameters

- Date Format: DD-MM-YYYY
- Time Format: HH:MM (24 Hours)
- IP Address
- OFP UID
- Data (Incoming & Outgoing):
    - Voltage
    - Current
    - Power Factor
    - Bi-Directional Energy